



Costs and Time Reduction Using Tap Thread Coated with Vanadium

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

Costs and Time Reduction Using Tap Thread Coated with Vanadium	0
1.0 Introduction	1
2.0 Best Practice Description.....	1
3.0 Implementation Steps	2
4.0 Benefits.....	3
5.0 Resources Required	3
6.0 Supporting Attachments / References	3
7.0 Related Best Practices	3
8.0 Acknowledgements.....	3

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1.0 Introduction

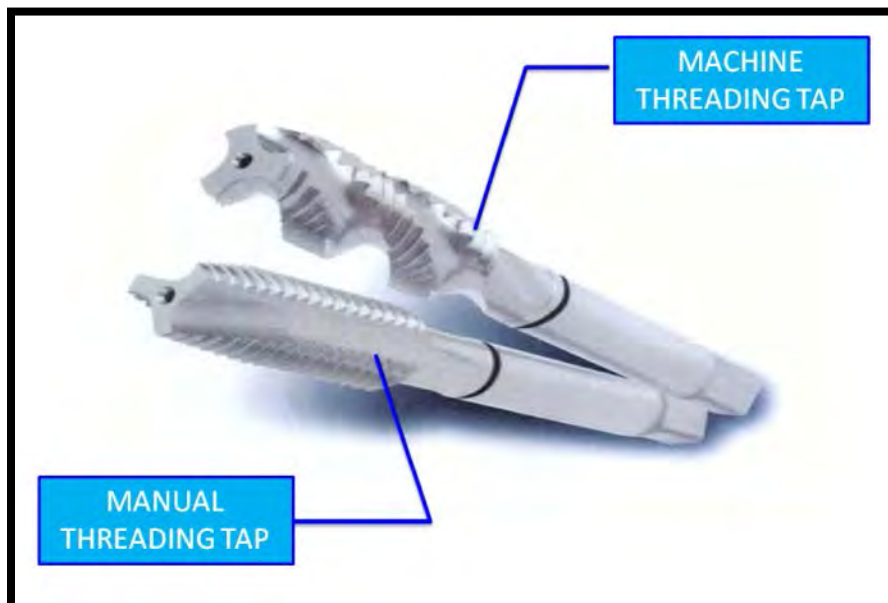
Improved threading tool allows a considerable reduction of time and costs in cleaning blocks, implemented in the machine shop of Ferreyros.

2.0 Best Practice Description

The focus of this Best Practice is to reduce the time and costs for cleaning waste such as grease, oxides, Lock-tite. This improvement occurs as a result of the high cost that generated use manuals HSS Taps Thread, increasing the cleaning time of blocks (See picture 1 and 2).



Picture 1. Cleaning thread



Picture 2. Tap identification

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DORMER -Threading tap - 1/2" x 13 UNC			OSG - Threading tap - 1/2" x 13 UNC		
Components	BLOCK N.	Holes	Components	BLOCK N.	Holes
Block 1	135323-01	103	Block 1	135821-01	98
Block 2	135318-01	24	Block 2	135923-01	146
Block 3	135248-01	125	Block 3	135821-01	44
Block 4	135358-01	100	Block 4		127
Block 5	135262-01	36	Block 5	135064-01	139
		388	Block 6	135759-01	147
			Block 7	136044-01	44
			Block 8	135241-01	122
			Block 9	136293-01	24
			Block 10	135968-01	75
			Block 11	136369-01	147
			Block 12	136288-01	147
					1260

Picture 3. Summary of results.

3.0 Implementation Steps

Before implementing this type of threading taps, cleaning time was almost twice that obtained with this implementation. (See picture 5)

We obtained a cost reduction of 34.83%.

➤ Reducing cleaning time thread					
Processing times - old					
Number of passes	Operation	Number of passes	Time (min)	Number of holes	Total time per operation (min)
1	Tapping 1	1	0.3	147	44.1
2	Tapping 2	1	0.3	147	44.1
					88.2 min
Total time per operation					88.2 min
Processing times - new					
Number of passes	Operation	Number of passes	Time (min)	Number of holes	Total time per operation (min)
1	Desbaste	1	0.3	147	44.1
					44.1 min
Total time per operation new					44.1 min

Picture 4. Reduction of time

➤ Reducing cost

Cost Reduction taps			
Make	Description	Unit	Unit Price
Dormer	Manual Threading tap 1/2"x13 UNC	1	58
OSG	Machine Threading tap 1/2"x13 UNC	1	37.8
Saving			\$ 20.2
Cost reduction			34.83%

Cost Reduction taps				
Make	Description	Unit	% / Using game taps for Block	Cost as use Block (3516)
Dormer	Manual Threading tap 1/2"x13 UNC	1	25.00%	\$14.5
OSG	Machine Threading tap 1/2"x13 UNC	1	7.70%	\$2.91

Picture 5. Cost reduction

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Costs and time reduction using tap thread coated with Vanadium

DATE
5/12/2015

CHG
NO
00

NUMBER
1501-4.02-1365

4.0 Benefits

- Reduced time in cleaning Block's by 50%.
- Cost reduction by 34.83%

5.0 Resources Required

- Threading OGS – EDP Number: 2832408

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

8.0 Acknowledgements

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